Post-Incident Analysis and SOC Metrics Report

Introduction

This practical exercise focused on performing post-incident analysis for a simulated phishing attack. The objective was to identify the root cause of the incident, document lessons learned, visualize contributing factors, and evaluate SOC performance using key metrics.

The analysis was conducted using structured methodologies such as the **5 Whys technique**, Fishbone Diagram, and SOC metrics like Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR).

Tools Used

Tool	Purpose
Google Sheets	Root cause analysis and documentation
SOC Metrics	Performance evaluation (MTTD, MTTR)

Incident Overview

A phishing email was delivered to a user, leading to unauthorized interaction with a malicious link. The incident resulted in potential credential exposure and required immediate response from the SOC team.

Root Cause Analysis (5 Whys Method)

The 5 Whys technique was used to identify the underlying cause of the incident.

Question	Answer
Why was the email opened?	User clicked malicious link
Why was the link clicked?	Weak email filtering
Why was filtering weak?	Lack of advanced phishing detection
Why was detection lacking?	Outdated security configurations
Why were configurations outdated?	Lack of regular security updates

Key Finding

The root cause of the incident was identified as **inadequate email security controls combined with outdated configurations**, allowing phishing emails to bypass detection.

Fishbone Diagram Analysis

Main Cause Categories Identified

- **People** → Lack of user awareness, insufficient training
- **Process** → No regular security audits, weak incident response procedures
- **Technology** → Ineffective email filtering, outdated detection systems
- **Environment** → Increased phishing attack volume

This visualization helped in systematically identifying all possible causes contributing to the incident.

SOC Metrics Calculation

To evaluate SOC performance, the following metrics were calculated:

- **Mean Time to Detect (MTTD)** = 2 hours
- **Mean Time to Respond (MTTR)** = 4 hours

Interpretation

- Detection was moderately delayed, indicating gaps in monitoring.
- Response time was acceptable but can be optimized with automation.

Metrics Summary

The SOC metrics indicate that the Mean Time to Detect (MTTD) was 2 hours, while the Mean Time to Respond (MTTR) was 4 hours. This suggests moderate detection efficiency but highlights the need for faster alert identification and improved automation to reduce response time and enhance overall incident handling effectiveness.